Practice Problems 1: System Modelling and Block Diagrams - Solutions

Question 1

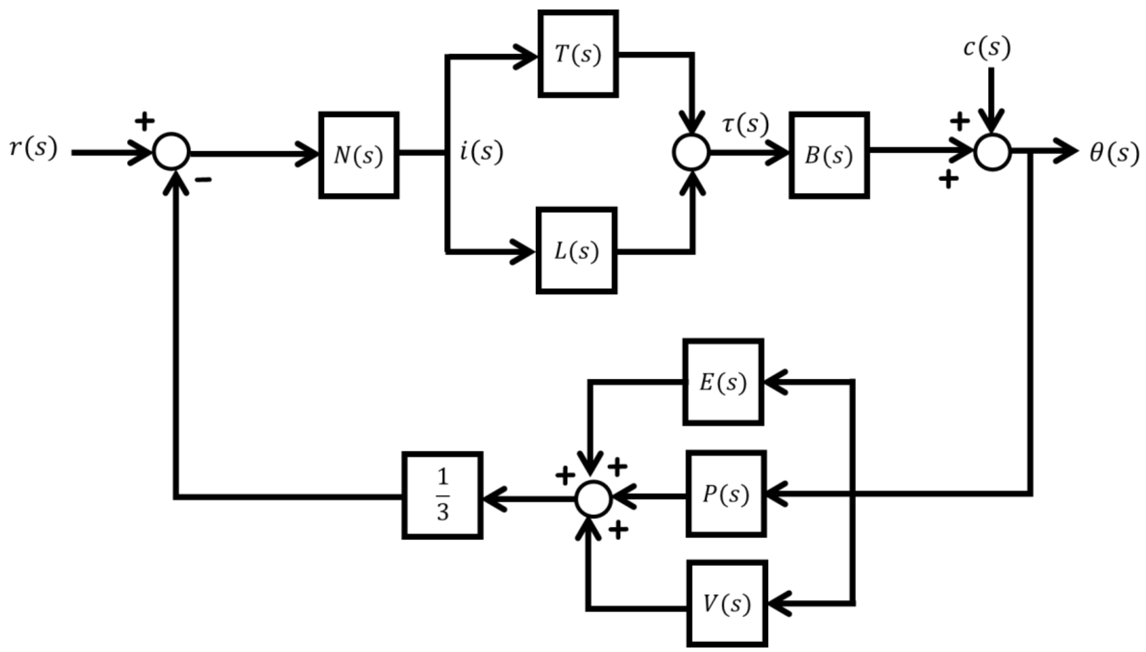
Neuromechanics is an area of biological research that studies how the nervous system controls the musculoskeletal system to bring about movement. A neuromechanics problem that the African Robotics Unit is interested in investigating is the role of the cheetah's tail in balancing its body. We have obtained footage from the Ann van Dyk Cheetah Centre of cheetahs being transported by bakkie. As the car bumps over the rough dirt roads around the sanctuary, the animals often lose their footing and are seen swinging their tails to recover.

a) Draw a block diagram modelling the cheetah in the vehicle. Here is the information your diagram must capture:

- The cheetah can be broken down into the following subsystems: the nervous system $N(s)$, the legs $L(s)$, the tail $T(s)$, the body $B(s)$, the eyes $E(s)$, the vestibular system $V(s)$ and the proprioceptive system $P(s)$.
- The cheetah is trying to control its body angle $\theta(s)$. It uses a combination of the eyes, vestibular system and proprioceptive system to detect this angle. You may consider the sensed angle to be the *average* of these three readings.
- The nervous system considers the difference between the desired angle $r(s)$ and the sensed position, and sends a nervous impulse $i(t)$ to the legs and tail to command them to respond.
- The legs and tail respond to the impulse, producing a combined torque $\tau(s)$ that acts to correct the body angle.
- The motion of the vehicle $c(s)$ also affects the angle of the body.

b) Reduce your block diagram into a transfer function relating the cheetah's body angle to (i) the desired angle, and (ii) the motion of the vehicle in terms of the given variables.

Solution:



b) Let's group the nervous system, tail and legs into a single system for the forward path, which we will call $F(s) = N(s)\big(T(s) + L(s)\big)B(s)$.

Similarly, we can group the sensory subsystems into a single block $S(s) = \frac{1}{s}\big(E(s) + P(s) + V(s)\big)$.

Moving around the loop gives the relationship $\theta = \big((r - \theta)F + c\big)S$,
which expands to $\theta = FSr - FS\theta + Sc$

From here, we get the transfer functions
(i) $\frac{FS}{1+FS}$
(ii) $\frac{S}{1+FS}$

You can substitute in the original blocks if you feel like it.

Question 2

Cape Town is a vitally important base for Antarctic research, as it is the closest port to the Antarctic circle. For this reason, marine robotics is one of the core research interests of the African Robotics Unit.

You wish to design a swarm of underwater robots to operate in the Southern ocean. To keep the robots working in a group, each one must be able to follow another robot at a set distance d.

a) Assuming the robots are moving in a straight line, use the information below to model the follower robot using a block diagram:

- The robot is propelled by an electric motor. Using a simplified model of the motor, we can relate the velocity of its shaft $\omega(t)$ to the input voltage $e(t)$ using the differential equation $J\dot{\omega}(t) = e(t) - B\omega(t)$, where J is the inertia of the motor and B is friction.
- The propellor converts the velocity of the motor to thrust. The thrust $F(t)$ can be assumed to be directly proportional to the shaft velocity, with proportionality constant ρ.
- The only two forces acting on the robot are the motor thrust, and hydrodynamic drag, which opposes the direction of motion. The robot has mass m. You can model the hydrodynamic drag as being proportional to the forward velocity $v(t)$, with proportionality constant D.
- The follower robot uses vision to determine the proximity to the leader, but this is not very reliable as murky water or poor lighting conditions can affect the result. You should assume the vision sensor *underestimates* the true distance by 10%.
- A controller applies a voltage to the motor based on the error between the desired distance d and the actual position $x(t)$. The controller is a PID, or *Proportional Integral Derivative* controller. Its output is a weighted sum of the error, its derivative and its time integral. The weights of these components are K, K_d and K_i respectively.
- Note: the output you are interested in is the position of the follower robot *relative* to the leader. Moving the follower forward will decrease this distance.

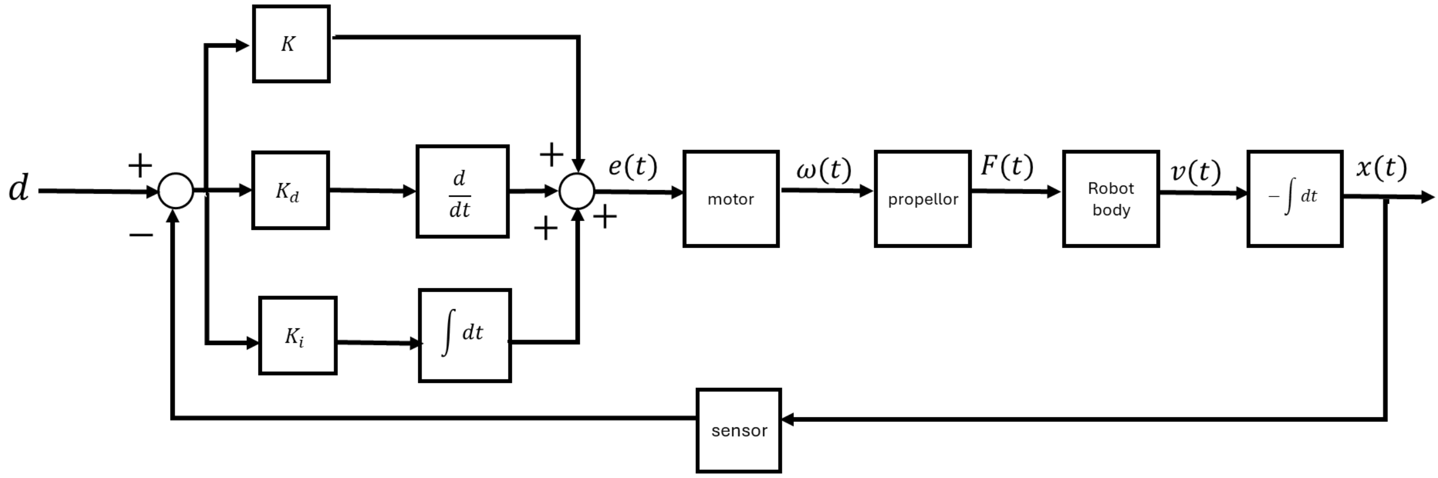
b) Based on your block diagram, devise the (i) open-loop and (ii) closed-loop transfer function for the robot in terms of the given variables and parameters.

Solution:

a) We are given information about the following subsystems, which can become the blocks in the model:

- The electric motor, which relates the input voltage $e(t)$ to the shaft velocity $\omega(t)$.
- The propellor, which relates $\omega(t)$ to the motor thrust $F(t)$.
- The robot body, which gains a forward velocity $v(t)$ as a result of the thrust $F(t)$. But we are interested in the position of the robot, so we will need to integrate this to get $x(t)$. The integration will also need to change the sign, as it represents a decrease in the relative distance between the robots.
- The controller, which is the sum of three subsystems: the proportional gain K, a differentiator with gain K_d and an integrator with gain K_i.
- The vision sensor that feeds back the position.

These blocks can be linked into the following diagram:



Transfer functions for these blocks can be determined as follows:

Controller: From the properties of the Laplace transform, the derivative is s and the integrator is $\frac{1}{s}$. The weights of each term are just scalars.

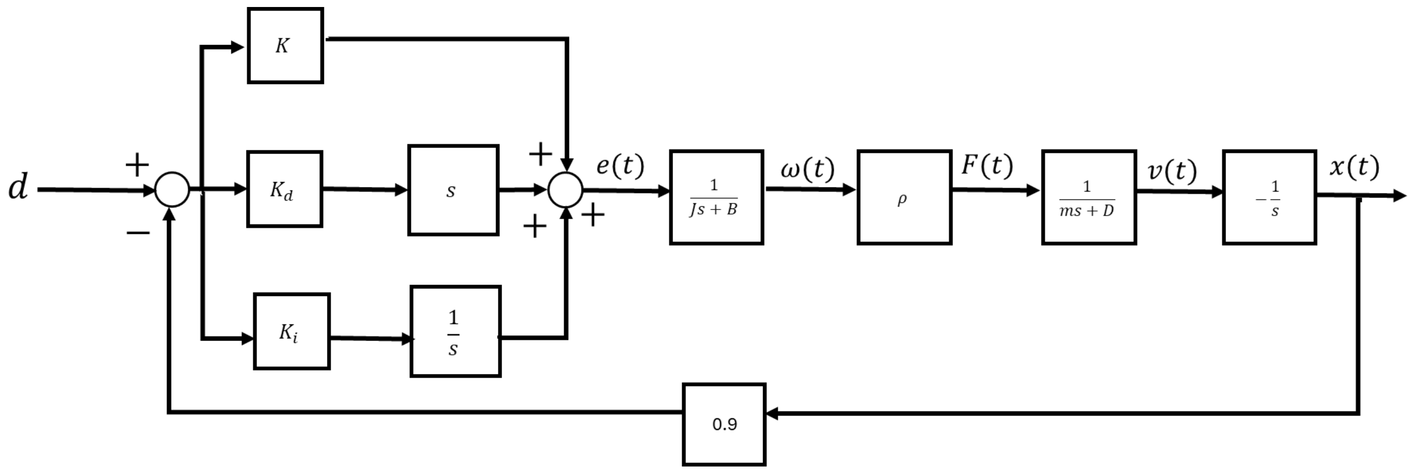
Motor: using the Laplace transform of the differential equation, we get $\frac{\omega(s)}{e(s)} = \frac{1}{B+Js}$

Propeller: this is just a scalar, ρ

Robot body: the differential equation $m\dot{v}(t) = F(t) - Dv(t)$ describes the motion of the robot through the water. Using the Laplace transform, this gives the transfer function $\frac{v(s)}{F(s)} = \frac{1}{D+ms}$.

Integrator: the transfer function of an integrator is $\frac{1}{s}$, so this will be $-\frac{1}{s}$

Sensor: the underestimation can be represented by a scalar 0.9



b) (i) The open loop model of the robot consists of the motor, propellor, body and integrator subsystems. These blocks are joined in cascade, so the model of the robot will be the product $G(s) = -\frac{\rho}{(Js+B)(ms+D)}$

(ii) The controller has the transfer function $C(s) = K + K_d s + K_i \frac{1}{s}$. The closed loop transfer function is $\frac{1}{1+0.9C(s)G(s)}$. The simplification from this point is tedious but trivial so let's leave it here.

Question 3

Feedforward control is a method where a known disturbance is predicted and preemptively counteracted. You want to use a combination of feedforward and feedback control to regulate the temperature in your tropical fishtank.

You have a heater $H(s)$, a thermometer $T_1(s)$ that measures the temperature inside the tank $x(s)$, and a thermometer $T_2(s)$ that measures the ambient temperature of the room $a(s)$.

A feedback controller $C(s)$ acts on the difference between the desired temperature $d(s)$ and the measured temperature in the tank.

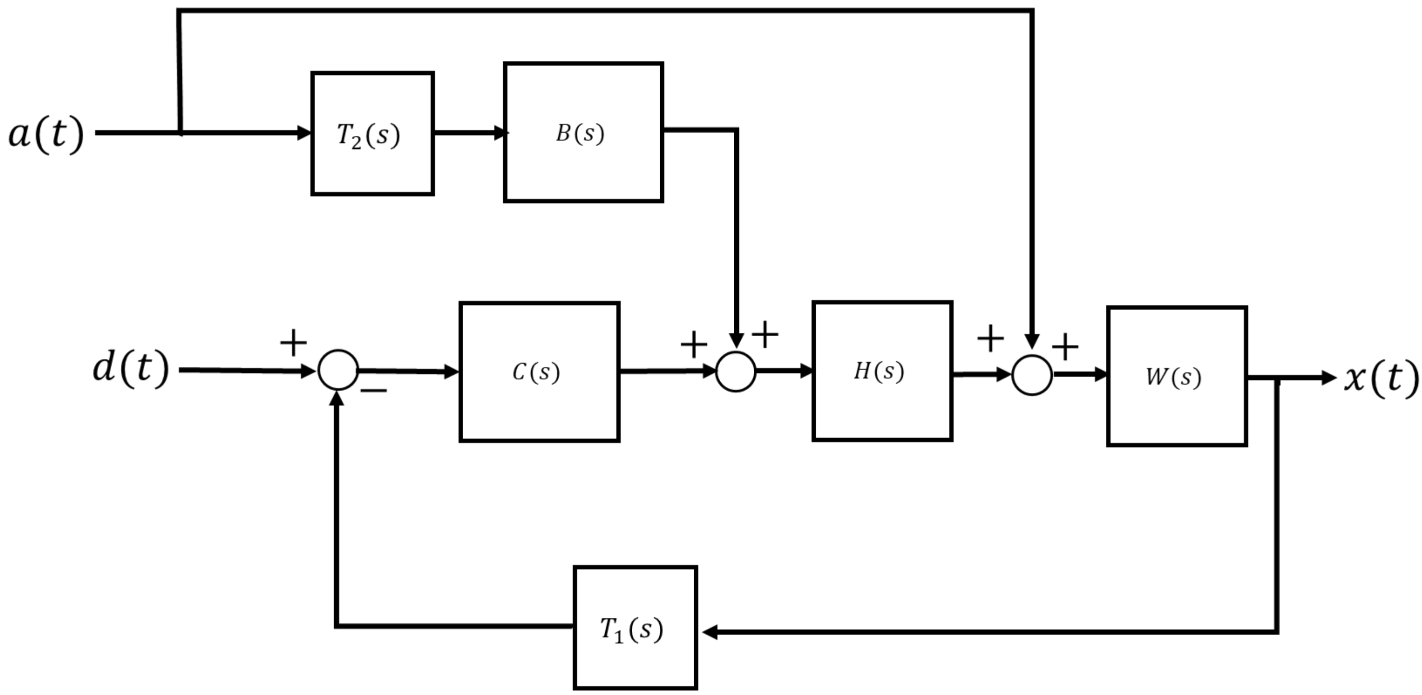
A feedforward controller $B(s)$ responds to the ambient temperature.

The outputs of both controllers are combined to give the input to the heater.

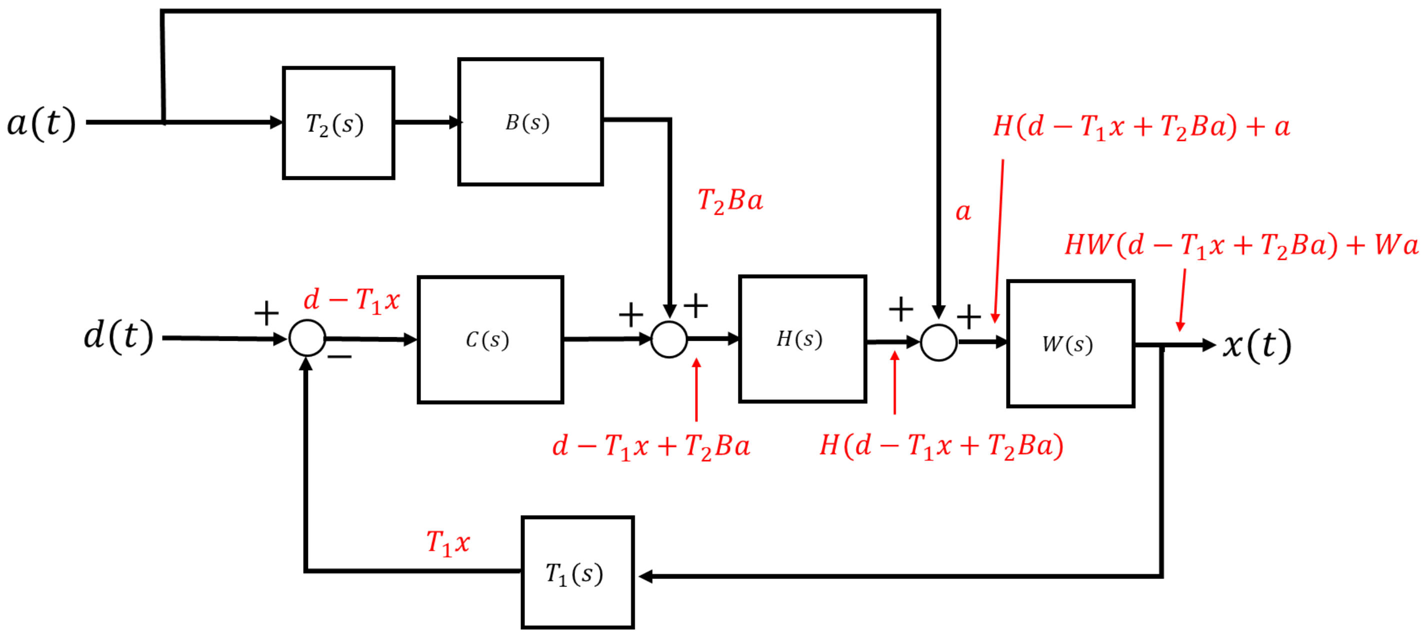
Draw a block diagram of this system and derive transfer functions relating (i) the desired temperature and (ii) the ambient temperature to $x(t)$, in terms of the given variables.

Solutions:

(a) Note: the ambient temperature is filtered through the feedforward controller, affecting the action of the heater, but it also acts on the tank directly.



b) This one is probably just easier to keep track of by annotating the block diagram at all the nodes.



From here, we can see that $x = HW(d - T_1x + T_2Ba) + Wa$
which expands to $x = HWd - HWT_1x + HWT_2Ba + Wa$

This gives the transfer functions:

(i) $\frac{HW}{1+HWT_1}$

(Note that it's just what we'd expect from the basic feedback system without the disturbance)

(ii) $\frac{HWT_2B+W}{1+HWT_1}$

Question 4

The Segway is a two-wheeled, self-balancing personal mobility device commonly used by security guards, tourists, people with trouble walking and Weird Al Yankovic. The Hoverboard is kind of the same thing, but smaller, and without a handle. (It doesn't hover.)

Figure 1: Weird Al on a Segway

What these two devices have in common is their mechanism of operation: tilting the deck forward causes the wheels to accelerate forwards, while tilting backwards causes them to slow down.

This is how the speed is controlled, but also how balance is achieved: to accelerate the wheels forwards, a positive forward torque must be applied. From Newton's Third Law, this means that an equal and opposite torque is applied to the board, causing it to rotate back towards its original flat position. Likewise, applying a negative torque on the wheels will counteract backward rotation of the board.

Clearly this is a rocking and rolling example of feedback control! Let's model it.

To model the Segway/ hoverboard, create a block diagram and then reduce it to get a closed-loop transfer function relating the desired upright position $r(s)$ to the angle of the rider $\theta(s)$ in relation to the vertical axis.

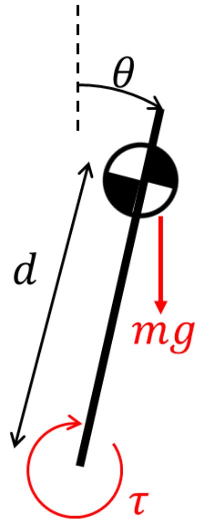


Figure 2: Inverted pendulum

Consider the following information:

- The rider and board can be modelled as an $inverted\ pendulum$ – a common model system in control engineering. This is a nonlinear system, but it can be simplified to a linear model in the vicinity of the upright equilibrium point when the angle θ is close to zero, as at this point, the $small\ angle\ theorem$ can be applied ($\sin(\theta) \approx \theta$).

  Besides the torque due to gravity and the input torque τ applied by the motor, you may assume that friction creates a damping effect proportional to the angular velocity of the board, $-\mu\dot{\theta}$.

  Devise a differential equation describing the motion of the inverted pendulum and hence, a transfer function for the rider block. Use the following symbols: m for the mass, J for the inertia, d for the distance from the pivot point (the board) to the centre of mass. (Assume initial rest.)

- The torque τ generated by the motor is proportional to the current i applied to its coil, with proportionality constant ρ.

- A gyroscopic sensor measures the *rate of change* of the angle i.e. $\dot{\theta}$

- A microprocessor reads the angular rate from the gyroscope and implements *proportional derivative* control of the angle. Note that it will need to integrate the rate to obtain the proportional component of the error. The proportional gain is K_p and the derivative gain is K_d.

Solution:

Rider: The torque acting on the rider due to gravity is created by the perpendicular component of the gravitational force, which is calculated as $mgd\sin(\theta) \approx mgd\theta$.

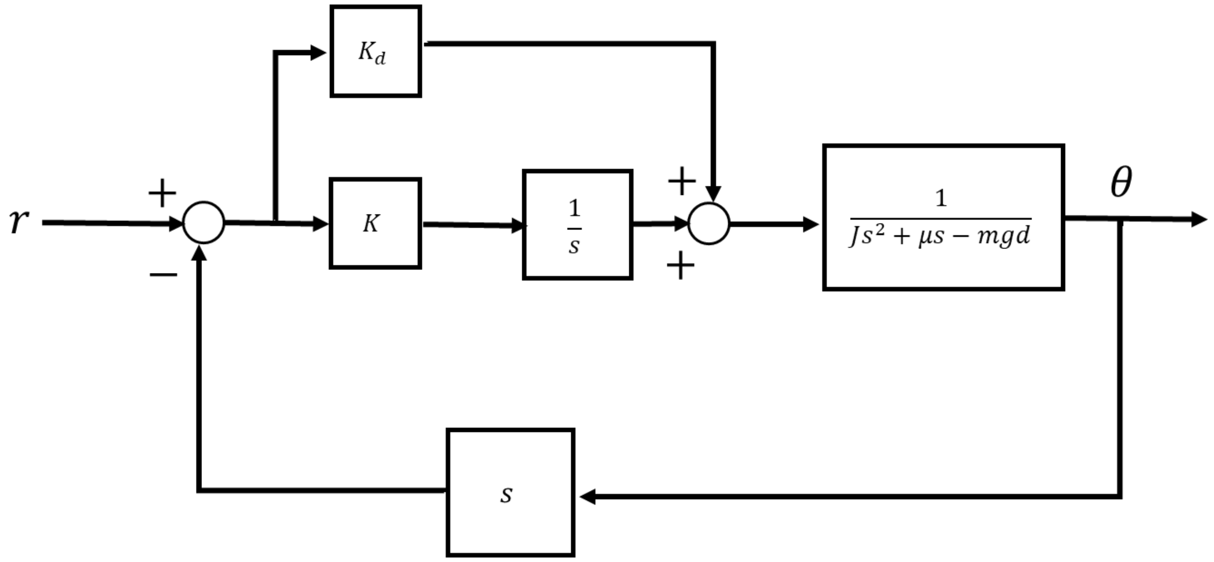
The differential equation is $J\ddot{\theta} = \tau - \mu\dot{\theta} + mgd\theta$

The inverse Laplace transform is $J\theta s^2 = \tau - \mu\theta s + mgd\theta$

This gives $G(s) = \frac{1}{Js^2+\mu s-mgd}$

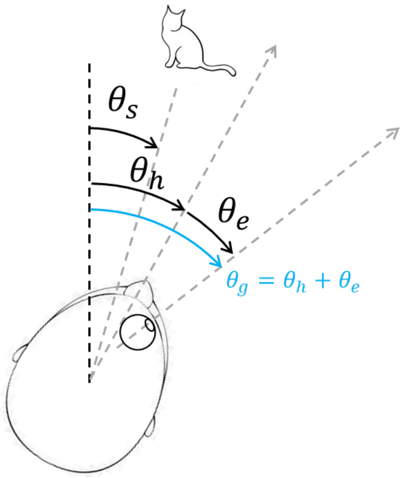
Sensor: The sensor reads the derivative of θ, so we can represent it with a pure differentiator s.

Controller: The PD controller can be constructed as in Question 2, but the output of the sensor is the velocity so it must be integrated to give the proportional component.



The closed loop transfer function is $\frac{CG}{1+CGH}$, where $C(s) = Kd + \frac{K}{s}$ is the controller transfer function and $H(s) = s$ is that of the sensor. You may simplify as needed.

Question 5



In humans and many animals, visual tracking is a combination of movement of the head and eyes. This is beneficial as the eyes are able to rotate quickly with much less effort than the relatively large and heavy head, allowing the gaze to keep up with fast-moving subjects.

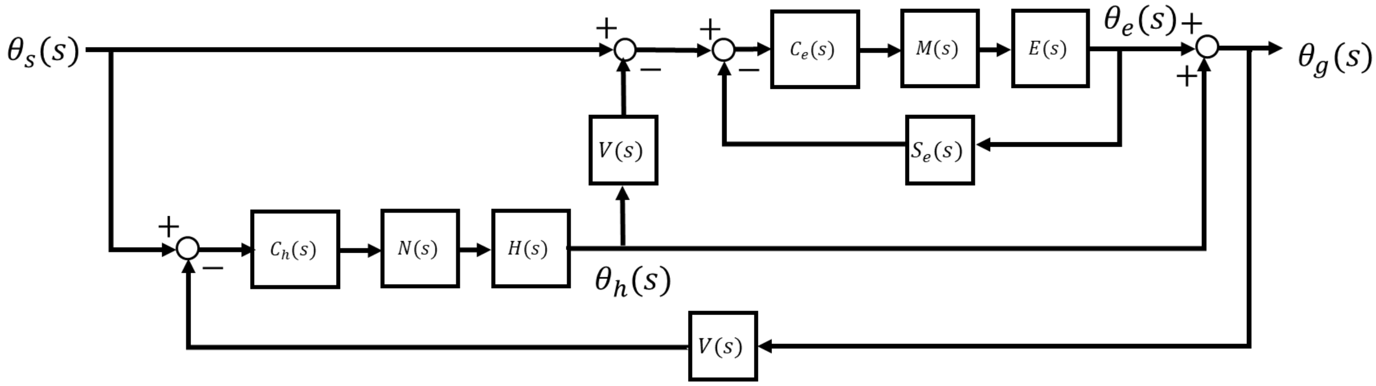
Draw a block diagram representing vision tracking based on the given information.

Hint: think of this control problem as having two loops: an inner loop controlling the position of the eyes relative to the head, and an outer loop controlling the position of the head. The difference between the gaze angle (sensed by the vision system, $V(s)$) and the angle to the subject causes the head to rotate towards the subject. Meanwhile, the eye will move within the head to make up the difference between the current head angle and the angle to the subject. The head sets the coarse direction of vision, while the eye makes the fine adjustments that allow gaze to lock onto the target.

- The gaze angle θ_g is a combination of the angle of the head θ_h and the angle of the eyes relative to the head θ_e. The angle to the subject θ_s is the target for the gaze angle.
- The desired angle for the eyes is the difference between the head angle angle (sensed by the vision system, $V(s)$), and the angle to the target. The eye angle is sensed by muscle spindles in the eye socket $S_e(s)$, and fed back to the neural controller of the eyes $C_e(s)$, which then sends impulses to the muscles of the eye socket $M(s)$. These produce a torque to rotate the eye $E(s)$.
- The nervous system sends impulses to the neck muscles $N(s)$, which produce torque to rotate the head $H(s)$. The neural controller of the head $C_h(s)$ responds to the difference between the gaze angle, sensed by $V(s)$, and the angle to the subject.

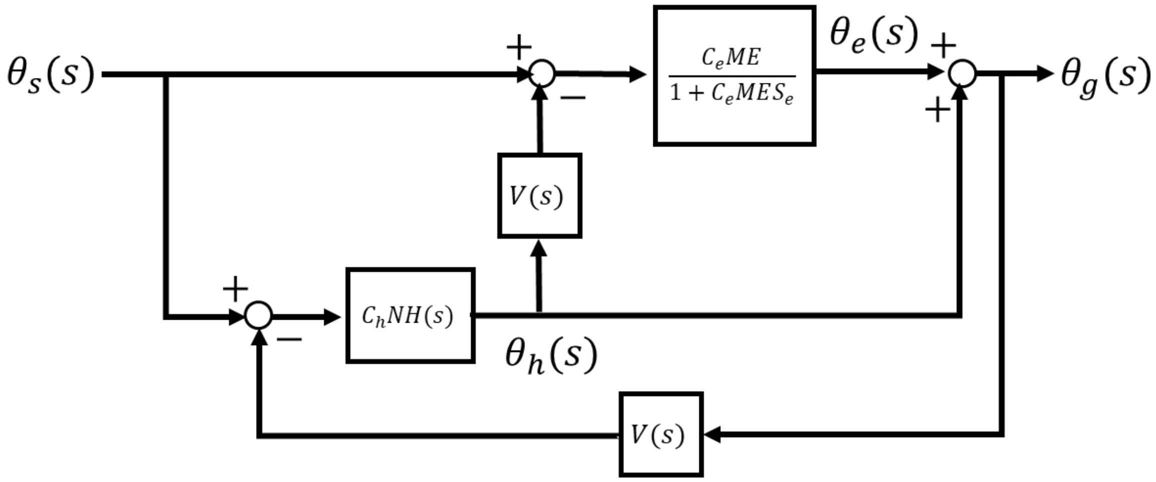
Based on your block diagram, devise a transfer function for visual tracking in terms of the given variables and subsystems.

Solution:

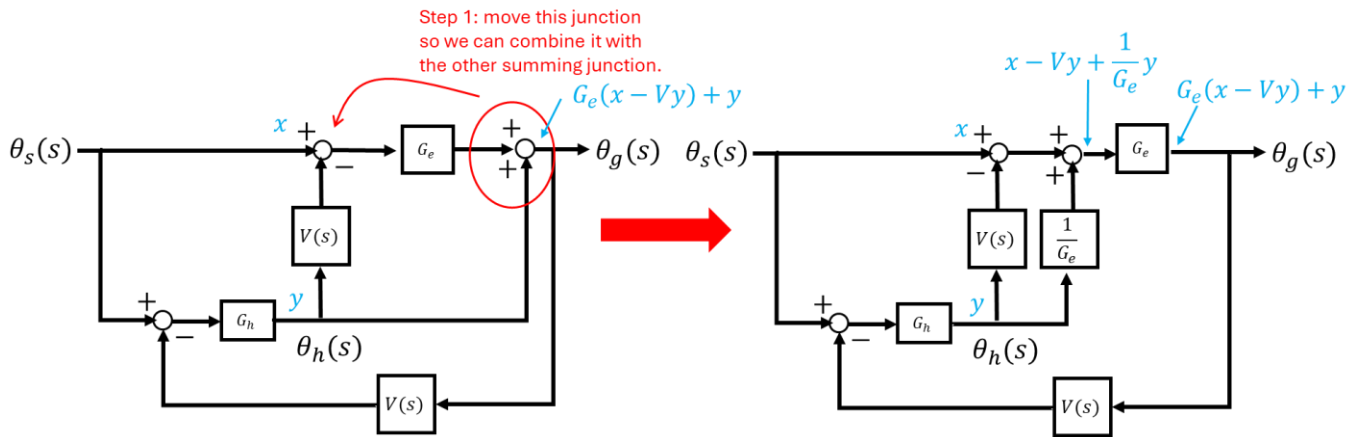


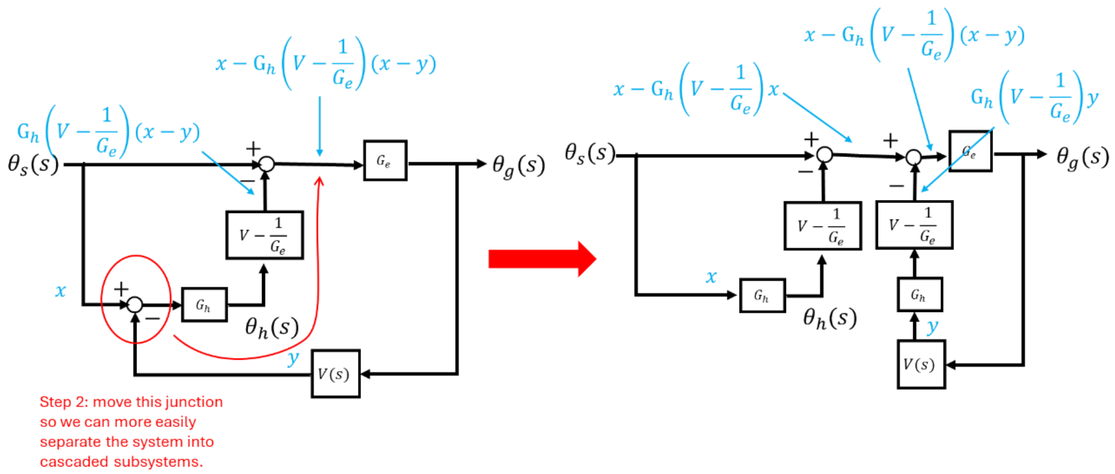
We can simplify the cascaded eye system (controller, muscles and eye) into CME, and the eye loop into $G_e = \frac{C_e ME}{1+C_e MES_e}$. The cascaded head system simplifies to $G_h = C_h NH$.

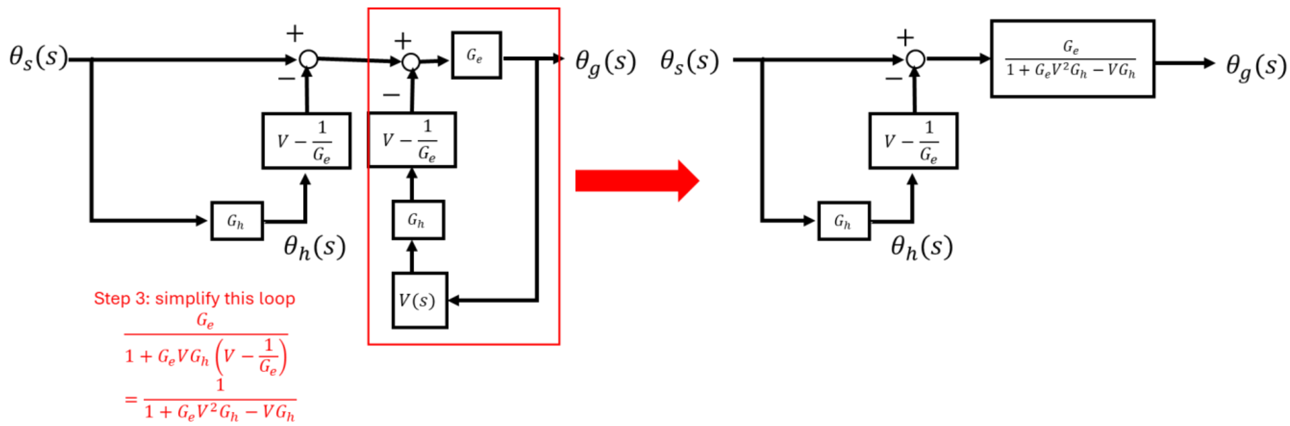
This reduces the block diagram to something that looks like this:

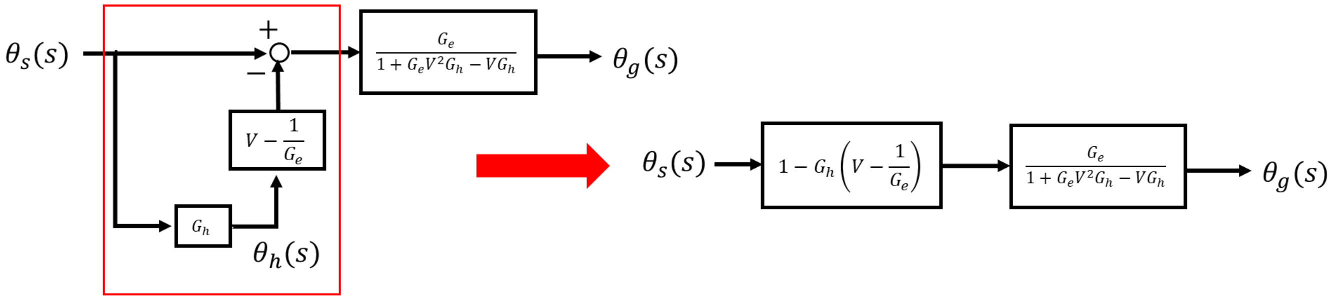


There are probably a few different ways to simplify this but here's how I went about it:









Step 4: simplify this parallel system

$$1 - G_h\left(V - \frac{1}{G_e}\right)$$

And from there you will simply multiply the cascaded systems together.